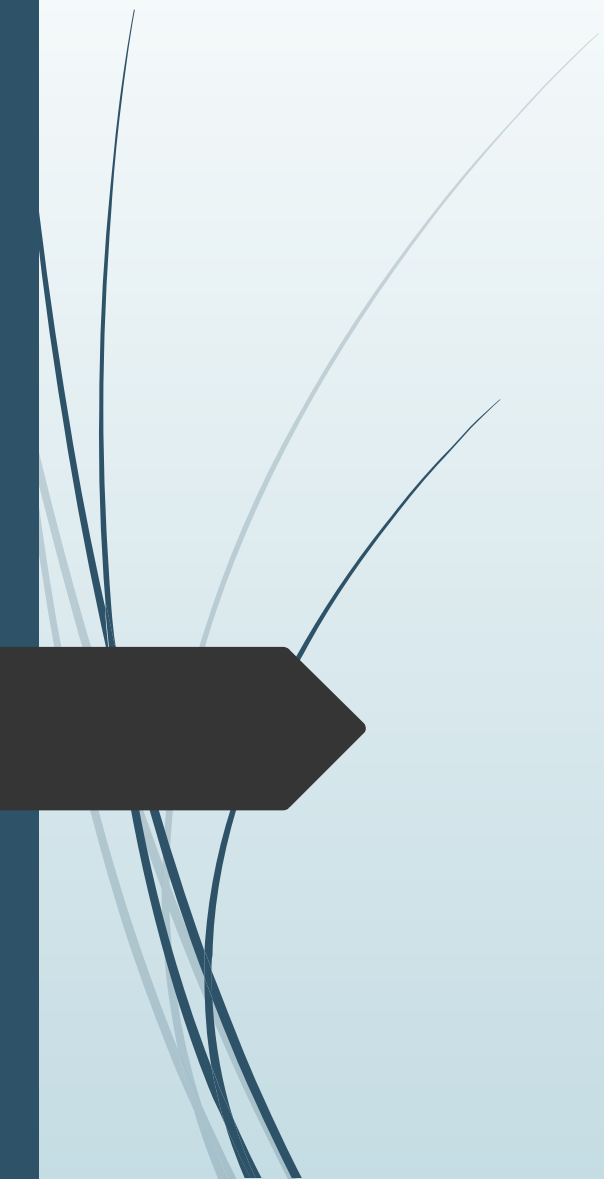


- Hough, *Method and means for recognizing complex patterns*, U.S. Patent No. 3,069,654, Dec 1962
- Line equation in Cartesian co-ordinates is:
  - $y = mx + c$ m is slope, c is y-intercept
- Rearranging it slightly, we get:
  - $c = (-x)m + y$ which for a specific point (x_i, y_i) becomes $c = (-x_i)m + y_i$
- This can be thought of as the equation of line in parameter space; i.e in the (m, c) coordinate system with slope $-x_i$ and c-intercept y
- Each point in parameter space is a model

- N samples needed to fit a model (2 points to fit a line)
- But even one sample brings some information
- In the space of all possible models, vote for ones that satisfy a given sample
- Collect votes for all samples, and seek for consensus

Hough Transform: Polar Parametrization

- The Cartesian formulation is problematic for vertical lines. Why?

- The Cartesian formulation is problematic for vertical lines. Why?
  - The slope is unbounded for vertical lines.
- Consider a polar parametrization of the line:
  - $\rho = x\cos\theta + y\sin\theta$
  - ρ is distance of line from origin and θ is angle made by normal to x-axis
  - For given line, $\rho \geq 0$ and $0 \leq \theta \leq 360°$ are bounded

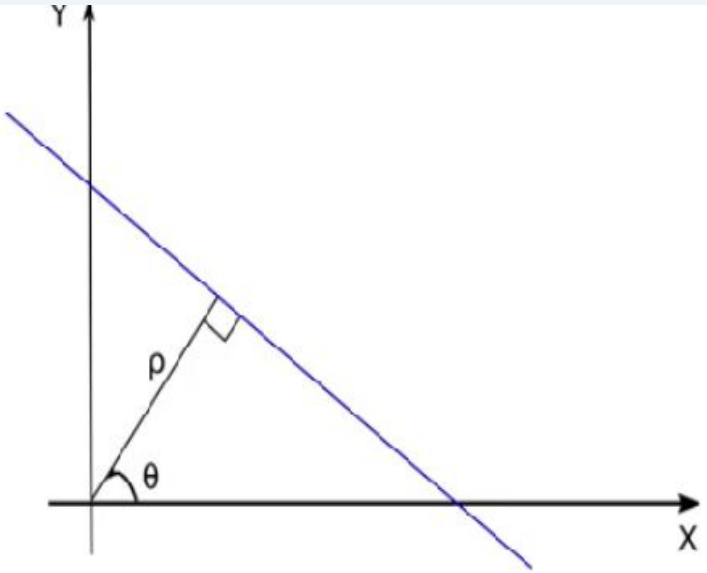


Figure 2: The ρ, θ parametrization.